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Artificial Bee Colony Improvement of Determinantal Sampling Designs

A Note on Horvitz–Thompson Scaled Auxiliary Variables

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Abstract:

We study an Artificial Bee Colony (ABC) search for improving fixed-size determinantal sampling designs with prescribed first-order inclusion probabilities. The reference design is Vincent's ordered real-valued P^π construction. Candidate designs are represented in the CaDSD parametrization by two matrices (ω, ρ) , which generate possibly complex Hermitian kernels. The first aim is to show that ABC can reduce the computed Horvitz–Thompson variance for the optimized variable z . The second aim is to explain why this improvement may not transfer to a target variable y , even when y and z are highly correlated: the variance depends on the scaled quantities y/π and z/π . A small simulation illustrates both the numerical gain for z and the instability caused by weak or negative transformed correlation.

Keywords: Artificial Bee Colony algorithm, determinantal sampling design, Horvitz–Thompson estimator, inclusion probabilities, auxiliary variable.

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1 Introduction

Let $\mathcal{U} = \{1, \dots, N\}$ be a finite population and let $\pi_k = \Pr(k \in S)$ be prescribed first-order inclusion probabilities, with $0 < \pi_k < 1$ and $\sum_{k=1}^N \pi_k = n$. For a variable $u = (u_1, \dots, u_N)^\top$, the Horvitz–Thompson estimator of the total $t_u = \sum_{k \in \mathcal{U}} u_k$ is

$$\hat{t}_{u, \text{HT}} = \sum_{k \in S} \frac{u_k}{\pi_k}.$$

The estimator is unbiased, but its variance depends on the dependence structure of the sampling design. This makes the construction of second-order inclusion probabilities a central problem in unequal-probability sampling.

Determinantal sampling designs provide a flexible way to control this dependence through a kernel matrix. If $S \sim \text{DSD}(K)$, then

$$\Pr(s \subseteq S) = \det(K_s), \quad s \subseteq \mathcal{U}.$$

For a Hermitian kernel, the first- and second-order inclusion probabilities are

$$\pi_k = K_{kk}, \quad \pi_{kl} = K_{kk}K_{ll} - |K_{kl}|^2, \quad k \neq l.$$

In the fixed-size case, K is a Hermitian projection kernel:

$$K^2 = K, \quad K = K^*, \quad \text{tr}(K) = n.$$

Thus, the diagonal fixes π , whereas the off-diagonal entries determine the repulsion between units and the variance of $\hat{t}_{u, \text{HT}}$.

Vincent’s ordered P^π construction gives a real-valued reference kernel with prescribed diagonal. For a variable u , the construction is used after ordering the population so that

$$\pi_1 \geq \dots \geq \pi_N, \quad \frac{u_1}{\pi_1} < \dots < \frac{u_N}{\pi_N}.$$

We write $V_{P^\pi}(u)$ for the corresponding reference variance. For any Hermitian candidate kernel K , the computed Horvitz–Thompson variance is

$$V_K(u) = u^* D_\pi^{-1} \{K \circ (I_N - \overline{K})\} D_\pi^{-1} u, \quad D_\pi = \text{diag}(\pi_1, \dots, \pi_N).$$

This note has two objectives. First, we use ABC to search for complex kernels that improve the Vincent reference for an optimized variable z . Second, we examine whether such an improvement is useful for a target variable y . This second question is delicate. Even if $\text{corr}(y, z)$ is high, the transformed correlation $\text{corr}(y/\pi, z/\pi)$ can be small or negative, and the latter quantity is more directly related to the Horvitz–Thompson variance.

2 Artificial Bee Colony search

ABC is a population-based metaheuristic inspired by the foraging behavior of honey-bee colonies [Karaboga \(2005\)](#); [Basturk and Karaboga \(2006\)](#); [Karaboga and Basturk \(2007\)](#). It has been developed mainly for continuous optimization and later extended and surveyed in many numerical contexts [Karaboga and Akay \(2009\)](#); [Akay and Karaboga \(2012\)](#); [Karaboga et al. \(2014\)](#). Here the food sources are not vectors of ordinary regression parameters; they are coordinates that define a determinantal kernel.

The search is performed in the CaDSD parametrization. A food source is

$$F = (\omega, \rho), \quad \omega \in [0, 1]^{n \times N}, \quad \rho \in [0, 1]^{n \times (N-1)}.$$

The CaDSD map sends F to a candidate kernel $K(F) = K(\omega, \rho)$. The Vincent reference point is represented by $\omega = 0$ and $\rho = 0.5$. In the experiments this point reproduces the real-valued P^π variance up to numerical tolerance, which checks that the ordering, diagonal, and variance calculation are aligned.

For a candidate F , the objective optimized by ABC is the efficiency ratio

$$E_z(F) = \frac{V_{P_z^\pi}(z)}{V_{K(F)}(z)}.$$

The search therefore maximizes $E_z(F)$, and $E_z(F) > 1$ means that the candidate kernel has smaller computed variance than the Vincent reference for the optimized variable. Feasibility is checked numerically by requiring

$$\max_k |K_{kk} - \pi_k| \leq \varepsilon_d, \quad -\varepsilon_e \leq \lambda_j(K) \leq 1 + \varepsilon_e, \quad |\text{tr}(K) - n| \leq \varepsilon_t,$$

for small tolerances $\varepsilon_d, \varepsilon_e, \varepsilon_t$. These checks are essential, because a strong value of the objective is meaningful only for a kernel that is close to a valid fixed-size determinantal design.

The ABC update follows the standard employed-bee, onlooker-bee and scout-bee structure. If F_i is the current food source and F_j is another source, a neighboring candidate is generated coordinatewise by

$$F'_{i\ell} = F_{i\ell} + \phi_{i\ell}(F_{i\ell} - F_{j\ell}), \quad \phi_{i\ell} \sim \text{Unif}[-1, 1],$$

followed by projection to the interval $[0, 1]$. In the employed-bee phase, each food source is locally perturbed and replaced if the new valid source improves E_z . In the onlooker-bee phase, food sources are selected with probability increasing in their fitness and are again locally perturbed. If a source fails to improve after a fixed number of trials, the scout-bee phase replaces it by a new random source. In our implementation the fitness is a monotone transform of E_z , so that larger efficiency gives larger selection probability.

Algorithm 1. *ABC search for a determinantal kernel.*

- *Initialize valid food sources $F_i = (\omega_i, \rho_i) \in [0, 1]^d$.*
- *For each source, compute $K(F_i)$, check validity, and evaluate $E_z(F_i)$.*
- *Apply employed-bee and onlooker-bee perturbations using the update above.*
- *Replace abandoned sources by random valid scout sources.*
- *Return the best valid F and the associated kernel $K(F)$.*

After optimizing for z , we also compute

$$E_y(F) = \frac{V_{P_y^\pi}(y)}{V_{K(F)}(y)},$$

where P_y^π is the Vincent reference built from the y/π ordering. This is not the optimized objective. It is a diagnostic for whether the z -optimized kernel is also useful for estimating the target variable y .

3 Simulation study

The experiment uses $N = 20$ and $n = 4$. Three cases are generated so that the raw correlations between the target variable y and the optimized variable z are approximately

$$\text{corr}(y, z) \in \{0, 0.80, 0.90\}.$$

For each case, ten inclusion-probability scenarios are considered: the equal-probability design and nine unequal-probability designs labelled $\pi_{0.10}, \dots, \pi_{0.90}$. In every scenario the population is ordered by z/π , so Vincent's monotonicity condition holds for the optimized variable.

The ABC run uses colony size 10, abandonment limit 20, and at most 1000 iterations. The search stops earlier if $E_z(F) \geq 3.5$. This threshold rule prevents the algorithm from running long after a practically meaningful improvement has already been obtained. The exported heatmaps keep the notebook labels for the three simulation cases; in the notation of this paper, the variable optimized by ABC is denoted by z , and the non-optimized target variable is denoted by y .

Figure 1 shows that ABC can improve the Vincent reference for the optimized variable. In 13 of the 30 scenarios, the threshold is reached before the maximum iteration limit, sometimes after only a few iterations. Other scenarios remain close to the reference value, indicating that the gain from complex CaDSD kernels depends on the joint structure of z and π .

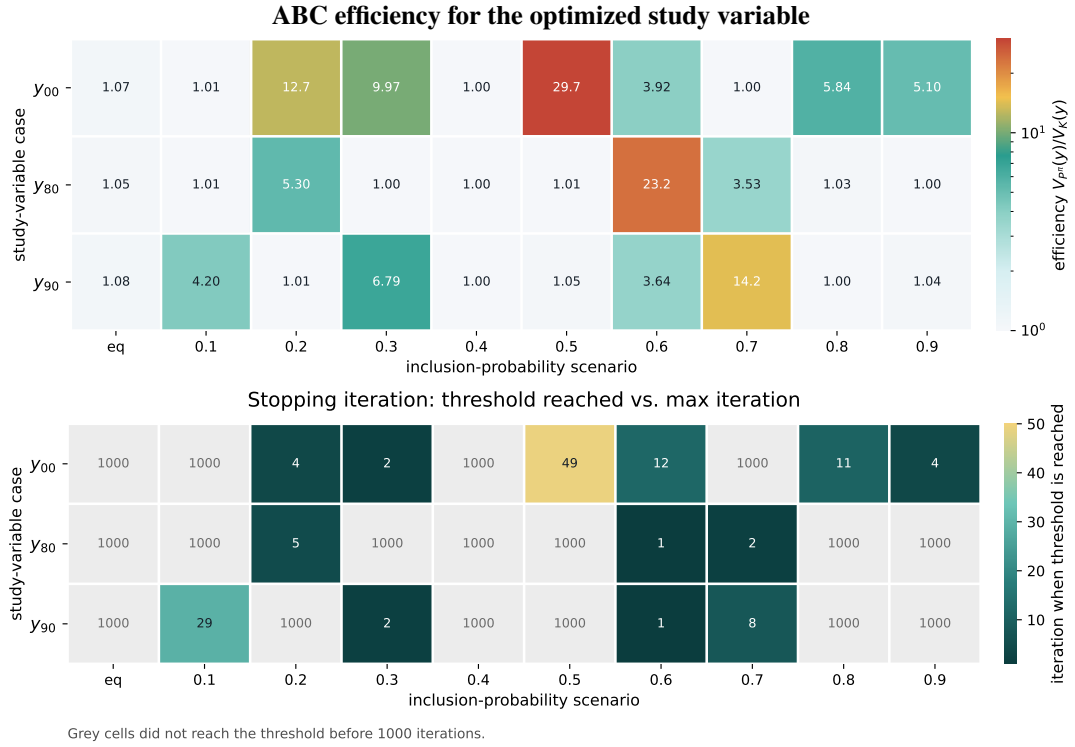


Figure 1: ABC performance for the optimized study variable, denoted z in the text. The upper panel reports $E_z(F) = V_{P_z^\pi}(z)/V_{K(F)}(z)$. The lower panel reports the stopping iteration; grey cells did not reach the threshold before 1000 iterations.

Figure 2 explains why a successful optimization for z does not automatically give a good design for y . Many values of E_y are below one. This happens even in rows where the raw correlation $\text{corr}(y, z)$ is high. The lower panel shows the reason: after division by the inclusion probabilities, $\text{corr}(y/\pi, z/\pi)$ may become weak or negative. Since the Horvitz–Thompson variance is driven by these scaled quantities, raw auxiliary correlation alone is not a reliable guide.

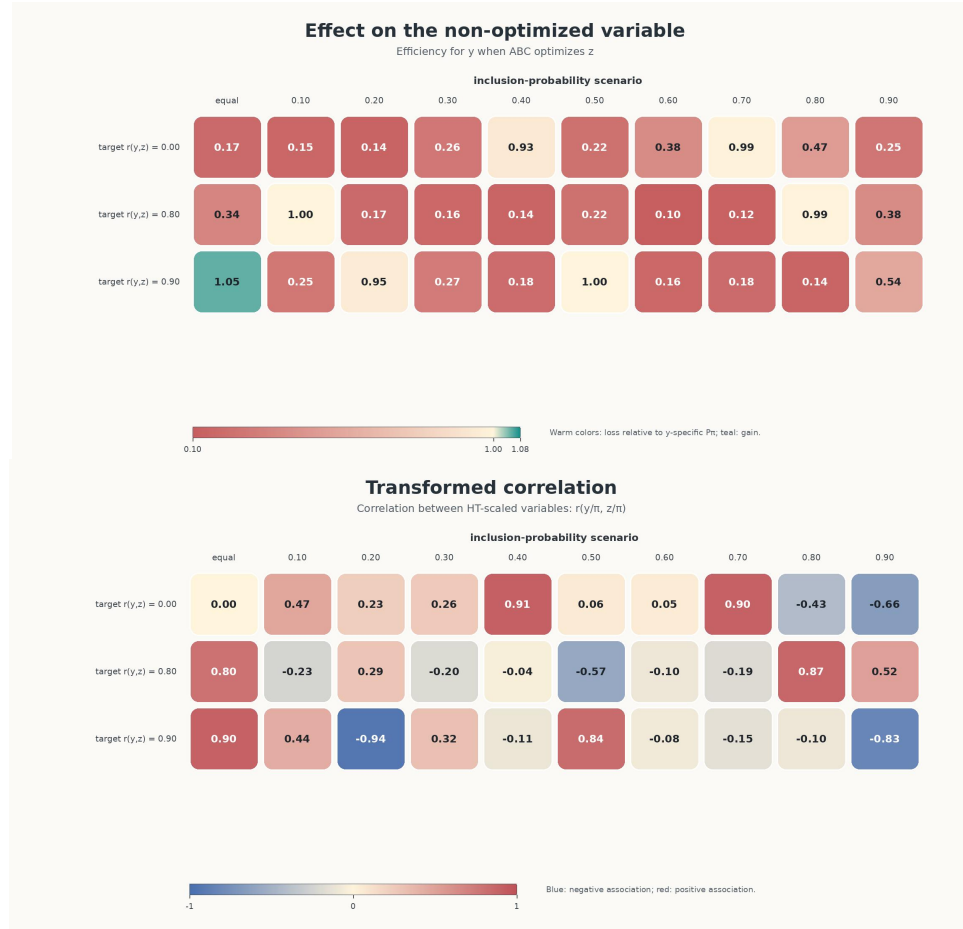


Figure 2: Transfer from the optimized variable z to the target variable y . The upper panel reports $E_y(F) = V_{P_y^\pi}(y)/V_{K(F)}(y)$. Values below one mean that the z -optimized kernel is worse for y than the Vincent reference based on the y/π ordering. The lower panel reports $\text{corr}(y/\pi, z/\pi)$.

Discussion and Conclusion

The experiment supports two messages. First, ABC is a useful numerical search tool for improving determinantal kernels within the CaDSD parametrization. Second, the improvement is target-specific: optimizing for z can be misleading for y when y/π and z/π are not aligned. The present results are numerical evidence rather than a final optimality theorem; the next step is independent verification of the exported kernels and larger real-data experiments.

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